Assignment1 – Probability

Given: April 22 Due: April 27

Problem 1.1 (Simple Sample Spaces)

In many important situations including all problems treated in this course, the sample space, probability measure, domains, and random variables can be given in a simplified form, namely by:

- a list of random variable declarations $X_1, ..., X_n$, each consisting of
  - a name such as X
  - a finite domain such as $D_X = \{0, 1, 2, 3\}$
- a probability function $P : \Omega \to [0,1]$ where $\Omega = D_{X_1} \times ... \times D_{X_n}$ such that $\sum_{e \in \Omega} P(e) = 1$

1. Define the corresponding probability space $\langle \Omega, P \rangle$

Solution: $P : \mathcal{P}(\Omega) \to [0,1]$ is defined by $P(A) = \sum_{e \in A} P(e)$. (Note this is a finite sum because all the D_X and thus Ω and A are finite.)

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2. Show that the satisfies the Kolmogorov axioms.

Solution: To show the Kolmogorov axioms:

- $P(\Omega) = \sum_{e \in \Omega} P(e) = 1$
- $P(\bigcup_i A_i) = \sum_{e \in \bigcup_i A_i} P(e) = \sum_i \sum_{e \in A_i} P(e) = \sum_i P(A_i)$. (because the A_i are pairwise disjoint.)

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3. Define the random variables $Y_1, ..., Y_n$ induced by the respective X_i.

Solution: For each X_i with domain D_{X_i}, we define a random variable $Y_i : \Omega \to D_{X_i}$ by $Y_i(x_1, ..., x_n) = x_i$ for $\langle x_1, ..., x_n \rangle \in \Omega$.

Problem 1.2 (Calculations)

Assume random variables X, Y both with domain $\{0, 1, 2\}$, whose joint probability distribution $P(X, Y)$ is given by

x	y	$P(X = x, Y = y)$
0	0	a
0	1	b
0	2	c
1	0	d
1	1	e
1	2	f
2	0	g
2	1	h
2	2	i

1. Give all subsets of the probabilities $\{a, b, c, d, e, f, g, h, i\}$ that sum to 1.

Solution: Only $\{a, b, c, d, e, f, g, h, i\}$

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2. In terms of $a, b, c, d, e, f, g, h, i$, give $P(X \neq 0)$.

Solution: $d + e + f + g + h + i$

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3. In terms of $a, b, c, d, e, f, g, h, i$, give $P(X = 1, Y = 0)$.

Solution: d

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4. In terms of $a, b, c, d, e, f, g, h, i$, give $P(X = 1 \mid Y = 0)$.

Solution: $d/(a + d + g)$

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5. In terms of $a, b, c, d, e, f, g, h, i$, give $P(X + Y = 2)$.

Solution: $c + e + g$

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6. In terms of $a, b, c, d, e, f, g, h, i$, give $P(X + Y = 2 \mid X > Y)$.

Solution: $g/(d + g + h)$

Problem 1.3 (Stochastic and Conditional independence)

Consider the following random variables:

- three flips C_1, C_2, and C_3 of the same fair coin, which can be heads or tails
- the variable E which is 1 if both C_1 and C_2 are heads and 0 otherwise
- the variable F which is 1 if both C_2 and C_3 are heads and 0 otherwise

Out of the above 5 random variables,

1. Give three random variables X, Y, Z such that X and Y are stochastically independent but not conditionally independent given Z,

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Solution: E.g., C_1 and C_2 with $Z = E$.

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2. Give three random variables X, Y, Z such that X and Y are not stochastically independent but conditionally independent given Z.

Solution: E.g., E and F with $Z = C_2$.

Problem 1.4 (Basic Probability)

Let A, B, C be Boolean random variables, and let a, b, c denote the atomic events that A, B, C, respectively, are true. Which of the following equalities are always true? Justify each of your answers in one sentence.

1. $P(b) = P(a, b) + P(\neg a, b)$

Solution: True (marginalization over A)

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2. $P(a) = P(a \mid b) + P(a \mid \neg b)$

Solution: Not true (e.g. $P(a \mid b) = P(a \mid \neg b) = 0.6$ would result in $P(a) = 1.2$)

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3. $P(a, b) = P(a) \cdot P(b)$

Solution: Not true (only true if A and B are stochastically independent)

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4. $P(a, b \mid c) \cdot P(c) = P(c, a \mid b) \cdot P(b)$

Solution: True (using product rule, both sides become $P(a, b, c)$)

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5. $P(a \vee b) = P(a) + P(b)$

Solution: Not true (general form is $P(a \vee b) = P(a) + P(b) - P(a, b)$)

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6. $P(a, \neg b) = (1 - P(b \mid a)) \cdot P(a)$

Solution: True ($1 - P(b \mid a) = P(\neg b \mid a)$ and via product rule we get $P(a, \neg b)$)

Problem 1.5 (Chained Production Elements)

An apparatus consists of six elements A, B, C, D, E, F. Assume the probabilities $P(b_X)$, that element X breaks down, are all stochastically independent, with $P(b_A) = 5\%$, $P(b_B) = 10\%$, $P(b_C) = 15\%$, $P(b_D) = 20\%$, $P(b_E) = 25\%$, and $P(b_F) = 30\%$.

Note: We deliberately differentiate between *not being operational* and *being broken*. If an element breaks, it is not operational; if an element is not operational, either it or the linked element broke.

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1. Assume the apparatus works if and only if at least A and B are operational, C and D are operational, or E and F are operational. What is the probability the apparatus works?

Solution: Let W be a random variable stating that the apparatus works. Let O_X be a random variable indicating that element X is operational.

In this problem, o_X is equivalent to $\neg b_X$ for all elements X.

$$\begin{aligned}
P(w) &= P(o_A \wedge o_B \vee o_C \wedge o_D \vee o_E \wedge o_F) \\
&= 1 - P(\underbrace{\neg(o_A \wedge o_B) \wedge \neg(o_C \wedge o_D) \wedge \neg(o_E \wedge o_F)}_{\text{all events are independent}}) \\
&= 1 - P(\neg(o_A \wedge o_B)) \cdot P(\neg(o_C \wedge o_D)) \cdot P(\neg(o_E \wedge o_F)) \\
&= 1 - P(\neg o_A \vee \neg o_B) \cdot P(\neg o_C \vee \neg o_D) \cdot P(\neg o_E \vee \neg o_F) \\
&= 1 - P(b_A \vee b_B) \cdot P(b_C \vee b_D) \cdot P(b_E \vee b_F) \\
&= 1 - (P(b_A) + P(b_B) - P(b_A) \cdot P(b_B)) \cdot (P(b_C) + P(b_D) - P(b_C) \cdot P(b_D)) \\
&\quad \cdot (P(b_E) + P(b_F) - P(b_E) \cdot P(b_F)) \\
&= 1 - (0.05 + 0.1 - (0.05 \cdot 0.1)) \cdot (0.15 + 0.2 - (0.15 \cdot 0.2)) \cdot (0.25 + 0.3 - (0.25 \cdot 0.3))
\end{aligned}$$

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2. Consider a different scenario, in which the elements A and C, D and F and B and E are pairwise linked; such that if either of them breaks down, then the linked element is not operational either. What is the probability that the apparatus works now?

Solution: Using the exclusion principle:

$$\begin{aligned}
P(w) &= P(o_A \wedge o_B \vee o_C \wedge o_D \vee o_E \wedge o_F) \\
&= P(o_A, o_B) + P(o_C, o_D) + P(o_E, o_F) - P(o_A, o_B, o_C, o_D) - P(o_A, o_B, o_E, o_F) - P(o_C, o_D, o_E, o_F) \\
&\quad + P(o_A, o_B, o_C, o_D, o_E, o_F)
\end{aligned}$$

Due to the links, o_X is equivalent to $\neg b_X \wedge \neg b_{\widetilde{X}}$ where $\widetilde{X}$ is the element that X is linked with. So, for example, o_A is equivalent to $\neg b_A \wedge \neg b_C$. This gives us

$$\begin{aligned}
P(w) &= \underbrace{P(\neg b_A, \neg b_C, \neg b_B, \neg b_E)}_{=P(o_A, o_B)} + P(\neg b_A, \neg b_C, \neg b_D, \neg b_F) + P(\neg b_B, \neg b_E, \neg b_D, \neg b_F) \\
&\quad - P(\neg b_A, \neg b_B, \neg b_C, \neg b_D, \neg b_E, \neg b_F) - P(\neg b_A, \neg b_B, \neg b_C, \neg b_D, \neg b_E, \neg b_F) \\
&\quad - P(\neg b_A, \neg b_B, \neg b_C, \neg b_D, \neg b_E, \neg b_F) + P(\neg b_A, \neg b_B, \neg b_C, \neg b_D, \neg b_E, \neg b_F) \\
&= P(\neg b_A, \neg b_C, \neg b_B, \neg b_E) + P(\neg b_A, \neg b_C, \neg b_D, \neg b_F) + P(\neg b_B, \neg b_E, \neg b_D, \neg b_F) \\
&\quad - 2P(\neg b_A, \neg b_B, \neg b_C, \neg b_D, \neg b_E, \neg b_F) \\
&= P(\neg b_A)P(\neg b_B)P(\neg b_C)P(\neg b_E) + P(\neg b_A)P(\neg b_C)P(\neg b_D)P(\neg b_F) + P(\neg b_B)P(\neg b_D)P(\neg b_E)P(\neg b_F) \\
&\quad - 2P(\neg b_A)P(\neg b_B)P(\neg b_C)P(\neg b_D)P(\neg b_E)P(\neg b_F) \\
&= 0.5450625 + 0.4522 + 0.378 - 2 \cdot 0.305235 \approx 76\%
\end{aligned}$$
